Parent-Level Debt and Subsidiary-Level Risk-Taking: Evidence from Large Bank Holding Companies *

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Abstract

We study how a bank holding company's (BHC) parent capital structure affects subsidiary bank risk-taking. Higher parent debt is associated with lower subsidiary bank risk-taking. Exploiting the U.S. introduction of single point of entry (SPOE) resolution and total loss-absorbing capacity (TLAC) requirements, we show that large non-GSIB BHCs unexpectedly exempted from parent-level debt requirements reduced parent debt, and their subsidiary banks increased risk-taking. A model in which parent debt is partly downstreamed as bank equity explains these findings: parent debt curbs subsidiary bank-level risk-taking while the parent remains solvent, but increases it once the parent would default alongside its subsidiary bank.

Keywords: bank holding company, parent-level debt, bank subsidiary, risk management

JEL Codes: G21, G28, G32, G33, G38

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1 Introduction

Since Jensen and Meckling [1976], corporate finance has a long tradition of considering the incentive implications of capital structure; the study of commercial banks makes no exception. Numerous empirical studies examine the impact of banks' capital structure on their risk-taking incentives, and virtually all studies explaining bank risk-taking would include bank capitalization as a control variable. Yet most modern commercial banks operate within a bank holding company (BHC). Avraham et al. [2012] provide an insightful overview of the organizational structure of US BHCs. Figure 1 reproduces their findings for a typical large BHC, in which the BHC simultaneously controls a bank subsidiary, and a non-bank subsidiary.[1] This multi-tiered organizational structure corresponds to a : the subsidiary bank's own, those of other subsidiaries, the parent company's, and the consolidated balance sheet of the whole group's. This multi-tiered capital structure raises an essential question: which entity's capital position matters? While prior research typically focuses on the capitalization of the commercial bank *or* that of the consolidated BHC, this paper argues that one needs to take a wholistic view, emphasizing the capital position of the parent company. Specifically, we study how the *parent debt ratio*, defined as the ratio of non-deposit debt issued by a BHC's parent company to the BHC's total assets, affects risk-taking by its commercial bank subsidiaries.

The parent debt ratio deserves attention for three reasons. First, parent debt is the typical funding source of subsidiary bank's equity: a parent that borrows and downstreams the proceeds as equity leaves its subsidiary banks well capitalised while the external debt at the BHC level increases. Neither the subsidiary bank's nor the consoli-

[1] Some BHCs also own foreign bank and non-bank subsidiaries.

dated BHC's balance sheet reveals this internal allocation. Second, post-crisis resolution policy has made parent debt the designated loss absorber of the global systemically important banks (GSIBs): under the single point of entry (SPOE) strategy and total loss-absorbing capacity (TLAC) requirements, losses incurred anywhere in the GSIBs are borne by the parent's creditors and shareholders. Third, because parent debt determines whether a subsidiary bank's losses fall on the subsidiary bank's own creditors or are absorbed inside the group, it directly shapes the limited-liability incentives that drive subsidiary bank's risk-taking.

The parent debt ratio shows significant time-series and cross-sectional variation: between 2011 and 2022, the average parent debt ratio rose from 0.72% to 1.50% (Figure 2). The sample in 2021 includes 251 BHCs. 50 of the 251 BHCs had no parent-issued debt, while in the top half of the distribution the average parent debt ratio was 2.63%. Despite such variation, whether parent debt affects the risk-taking of the subsidiary banks that hold most of the group's assets has not, to our knowledge, been examined. We address this gap by examining how the parent debt ratio influences the subsidiary bank' risk-taking—an important question given that commercial banks account for the bulk of BHC assets, rising from 89.03% in 2012Q1 to 94.45% in 2022Q4 (Figure 3). Non-bank subsidiaries enter our analysis only through the model, where their equity determines whether the parent can service its own debt when the subsidiary bank fails.

Theoretically, external debt issued at the parent level may affect the risk-taking of subsidiary banks through two opposing channels. The first is a ***loss-absorption (commitment) channel***. When the parent company raises external debt and downstreams the proceeds to its subsidiary banks as equity investment, the subsidiary bank's capital base expands. The resulting increase in capital constrains the subsidiary banks' incentives to take on risk, since potential losses are more fully absorbed by this equity

buffer. Therefore, this parent-level debt effectively serves as a contractual commitment that helps mitigate risk-taking at the subsidiary level. In contrast, if the parent company issues less debt, subsidiary banks may need to rely more heavily on their own debt issuance to finance investments, which can encourage greater risk-taking due to limited liability. Under limited liability, downside losses are partly borne by debt holders, while shareholders retain the upside gains.

The second is a ***BHC-level risk-shifting channel***. Parent-level debt raises the leverage of the BHC as a whole. Once the parent itself faces a non-trivial probability of default, limited liability at the BHC level shields its shareholders from downside losses, which are transferred to parent creditors. Additional parent debt may therefore amplify, rather than curb, risk-taking incentives across the organization. Moreover, servicing parent-level debt may require the subsidiary bank to upstream dividends, which can push it toward higher-yielding, and hence riskier, assets.

Which channel dominates is ultimately an empirical question. We therefore test these two mechanisms by examining the impact of the parent debt ratio on subsidiary banks' risk-taking. We combine U.S. bank-level regulatory data from the Call Reports with consolidated and parent-only BHC data from the FR Y-9C and FR Y-9LP. We measure subsidiary banks' risk-taking on both sides of the balance sheet. On the asset side, the loan and lease yield captures the ex ante riskiness of new lending, and the provision ratio captures the subsidiary bank's own forward-looking estimate of credit losses. On the liability side, leverage captures the extent to which the bank funds itself with its own debt, which is the channel through which limited liability operates. The Z-score combines both sides into a summary measure of distance to insolvency. Because most subsidiary banks are not publicly traded, these accounting-based measures are the natural choice.

We find that a higher parent debt ratio is associated with lower risk-taking by subsidiary banks. Specifically, an increase in the parent debt ratio is correlated with lower loan yields, lower provision ratios, and higher Z-scores at the subsidiary bank level. These findings suggest that a higher parent debt ratio reduces risk-taking within subsidiary banks.

While the empirical evidence is suggestive, a key concern in identifying the effect of parent debt on subsidiary banks' risk-taking is omitted variable bias. Specifically, when a subsidiary bank faces stringent regulatory capital requirements, it may reduce its risk-taking to avoid potential capital losses. At the same time, the parent BHC may issue additional external debt and inject the proceeds into the subsidiary bank to help it meet these requirements. In this case, the observed negative relationship could be driven by the omitted factor, namely the tightening of regulatory capital constraints, rather than by a causal effect of parent debt ratio.

To address this identification concern, we exploit the introduction of the SPOE resolution strategy and the TLAC requirement in the United States as a natural experiment. These rules unexpectedly exempted BHCs that were subject to the enhanced prudential standards (EPS) of the Dodd–Frank Act but were not classified as global systemically important banks (GSIBs). This exemption allowed large non-GSIB BHCs to sharply reduce their parent-level debt ratios, while the GSIBs' parent-level debt ratios remained elevated because of regulatory constraints. Exploiting this regulatory change, we conduct a difference-in-differences analysis with the large non-GSIB BHCs as the treatment group and BHCs below the EPS threshold as the control group. Importantly, we also use the GSIBs as a placebo treatment group.

Our treatment group, large non-GSIB BHCs, has received little attention in the literature. Empirical studies of resolution reform have focused on the covered firms: Berndt

et al. [2025] estimate market-implied bailout probabilities for U.S. GSIBs, Afonso et al. [2018] compare the ratings and spreads of parent companies and their bank subsidiaries at the four largest BHCs after the SPOE announcement, and Lewrick et al. [2019] study the pricing of bail-in bonds issued by GSIBs and other large banks. Theoretical work likewise centers on the resolution of global banks [Bolton and Oehmke, 2019]. For our question, focusing on the exempted group has two advantages. First, both the GSIBs and the large non-GSIB BHCs were subject to the EPS, so the two groups faced a common set of regulations, with the exception of TLAC, from which the large non-GSIB BHCs were unexpectedly exempted. The exemption therefore changed the expected parent-debt requirement of the large non-GSIB BHCs without altering the other rules they faced. Second, the GSIBs, which share the complexity and supervisory regime of the exempted large non-GSIB BHCs and are closest to them in size, can serve as a comparison group rather than as the object of study.

We find consistent evidence across risk-taking measures that supports the hypothesis that a higher parent debt ratio lowers subsidiary banks' risk-taking. By contrast, we find no comparable change for the GSIB placebo group, which alleviates concerns that our findings are driven by confounding factors, especially those related to BHC size. These results are also consistent with our baseline analysis of nearly 300 U.S. BHCs from 2011 to 2021Q2, which controls for bank- and BHC-level characteristics and includes subsidiary-bank and quarter fixed effects.

To interpret these findings, we develop a stylized model of a BHC in which a parent company owns a subsidiary bank and a non-bank subsidiary. The parent is financed with outside equity and external debt and downstreams a fraction of its funds to the subsidiary bank as equity. Holding the size of the subsidiary bank's balance sheet fixed, each additional unit of parent debt therefore partially displaces subsidiary bank de-

posits. Deposits are socially valuable because they provide a convenience yield, yet limited liability at the subsidiary bank level induces the parent's shareholders to choose excessive risk. The model generates a non-monotonic relationship between parent debt and subsidiary bank risk-taking. When the parent remains solvent in the event of subsidiary bank failure, additional parent debt reduces subsidiary bank leverage and forces the parent's shareholders to internalize a larger share of the subsidiary bank's losses, thereby mitigating risk-shifting. When parent debt exceeds the parent's investment in the non-bank subsidiary, however, the parent defaults whenever the subsidiary bank fails, and additional parent debt exacerbates risk-taking by introducing a second layer of limited liability at the holding company level. Subsidiary bank risk is thus minimized at the parent's solvency threshold. A constrained planner who chooses the BHC's capital structure but cannot control its risk choice, sets parent debt to trade off the reduction in subsidiary bank risk-taking against the forgone convenience yield of deposits, and optimally keeps parent debt at or below the solvency threshold. The model further predicts that a higher convenience yield on deposits is associated with lower parent debt and greater subsidiary bank risk-taking. Our empirical evidence that parent debt reduces subsidiary bank risk-taking is consistent with BHCs operating below the solvency threshold, the region in which the constrained social optimum lies.

Our paper highlights the function of internal capital markets within BHCs. Unlike existing studies—some of which focus on risk-sharing through internal capital markets among global banks (e.g., Cetorelli and Goldberg [2012])—we emphasize the role of internal capital markets in shaping the risk-taking incentives of domestic subsidiary banks within U.S. BHCs.

This study contributes to the large literature on the financial ratio regulations aimed at curbing financial institutions' risk-taking incentives. The specific ratio examined here

is the parent company's capital structure, which differs from prior research that primarily focus on capital structure at the commercial bank level or the consolidated BHC level (e.g., Keeley [1990], Shrieves and Dahl [1992], Demsetz and Strahan [1997], and Ellul and Yerramilli [2013]).

This paper also contributes to the literature on the organizational structure of financial institutions and risk-taking. Our analysis focuses on the multi-level organizational structures of BHCs and their distinct balance sheets—particularly that of the parent company—rather than the single balance sheet of integrated conglomerates emphasized by Freixas et al. [2007]. Moreover, we highlight the importance of considering existing BHC organizational structures when assessing risk-taking, in contrast to Kahn and Winton [2004] and Luciano and Wihlborg [2018], who design optimal structures to mitigate risk-taking. Closest to our approach is Li et al. [2024], who argue that a bank's organizational structure should be evaluated alongside its capital structure and bankers' compensation contracts when discouraging banker's risk-taking.

Our paper differs from the SPOE/TLAC literature (e.g., Bolton and Oehmke [2019], Clayton and Schaab [2025]) in two respects. First, that literature studies how parent-level loss-absorbing debt should be designed and deployed to resolve a bank once distress has occurred, whereas we study its ex-ante effect on incentives, before any distress materialises. Second, much of the SPOE literature is set in cross-border banking, where resolution is further complicated because national authorities may ring-fence local subsidiaries and subsidiaries have access to national safety-net subsidies. Our setting, in both the empirical analysis and the model, is purely domestic and abstracts from this additional layer of distortion.

Finally, we add to the nascent literature on parent debt in BHCs, where existing research is scarce. Tucker et al. [2023] show that the parent-only leverage ratio is positively

associated with the parent's credit default swap premium, suggesting that parent-only balance sheet data can inform assessments of parent credit risk and that consolidated reporting can obscure critical entity-level details. Building on these insights, we examine how parent company debt affects subsidiary bank risk-taking, offering a novel perspective on risk transmission within BHCs.

The paper is structured as follows: Section 2 describes the data sources, key variables, and summary statistics. Section 3 conducts the empirical analysis on the relationship between parent debt and subsidiary banks' risk-taking. Section 4 presents the difference-in-differences analysis. Section 5 introduces the theoretical model. Finally, Section 6 concludes.

2 Data and summary statistics

In this section, we describe our data sample, outline the construction of the parent debt ratio and subsidiary banks' risk-taking measures, and summarize our sample. Table 1 provides detailed definition of these variables.

2.1 Data sources

2.1.1 BHC-level data

Our BHC-level data are sourced from the Consolidated Financial Statements for Holding Companies (FR-Y9C) and Parent Company Only Financial Statements for Large Holding Companies (FR-Y9LP) provided by the Federal Reserve Bank. We use data spanning from 2011 to 2021Q2, which include quarterly observations of the income statement and

balance sheets of all U.S. BHCs and their parent-only companies. We drop BHCs that do not file the FR-Y9C or FR-Y9LP for each quarter during the entire sample period to avoid the impact of BHCs' joining or exiting on our empirical results. Finally, we collect data for 298 BHCs over the sample period.

2.1.2 Subsidiary bank-level data

The subsidiary bank-level data are from the U.S. Call Reports provided by the Federal Reserve Bank, covering the period from 2011 to 2022Q2. This dataset contains quarterly observations of the income statement and balance sheets of all U.S. commercial banks, some of which are the subsidiaries of BHCs. The data contain bank-level identifier (RSSD9348) that can be used to link to the BHC-level data. Finally, our balanced panel consists of 579 unique subsidiary banks.

2.2 Parent debt ratio

Our key explanatory variable is the parent debt ratio, defined as the parent's liability scaled by the BHC's total consolidated assets. The parent's liability includes the parent company's deposits, securities sold under agreements to repurchase, borrowings of commercial paper and other borrowings with a remaining maturity of one year or less, other borrowed money with a remaining maturity of more than one year, and subordinated notes and debentures.

2.3 Subsidiary bank's risk-taking

We measure subsidiary bank risk along both the asset and liability sides of the balance sheet. Our first measure is the Z-score, defined as the sum of the return on assets (ROA) and the equity-to-assets ratio, scaled by the standard deviation of ROA, where the equity-to-assets ratio is book equity divided by total assets. The Z-score thus captures both asset-side risk, through the level and volatility of returns, and liability-side risk, through capitalization; a higher Z-score indicates greater bank stability. Because the Z-score is highly skewed, we use its natural logarithm, which is approximately normally distributed (Laeven and Levine [2007]). Second, similar with Acharya et al. [2017], we measure leverage as the ratio of the subsidiary bank's debt to its equity. Third, we use two measures that capture risk-taking in the loan portfolio more directly. The loan and lease yield, defined as quarterly interest income on loans and leases divided by quarterly average loans and annualized, is an ex ante measure: higher yields indicate lending to riskier borrowers. The provision ratio, defined as provisions for credit losses divided by quarterly average loans and leases and annualized, reflects expected credit losses in the loan portfolio (Hirtle et al. [2020]).

2.4 Control variables

We use market data from the Center for Research in Security Prices (CRSP) and regulatory accounting data from Call Reports, FR Y-9C, and FR Y-9LP to construct subsidiary bank- and BHC-level controls that are considered as important determinants of a bank's risk-taking behavior.

These control variables include subsidiary bank size ($LnTA$), the subsidiary bank's leverage ratio ($Debt/Equity$), high-quality liquid assets ($HQLA$), and the BHC's Tobin's Q ($TobinQ$). Keeley [1990] shows that Tobin Q, as a proxy of market power or charter value, is negatively related to a bank's risk-taking. Boyd and Runkle [1993] suggest a negative relationship between bank size and bank risk. Acosta-Smith et al. [2020] find that a leverage ratio requirement associated with the introduction of the Basel III Leverage Ratio can incentivize banks bound by it to increase their risk-taking.

2.5 Summary statistics

Table 2, Panel A summarizes the sample statistics for the 2012-2021Q2 period. For the BHC-level variables, the parent debt ratio averages 1.87%. The average Tobin's Q for BHCs is 1.03. The mean value of HQLA ratio is 15.42%.

For the subsidiary bank-level variables, the mean logarithm of asset size is 15.02, which translates to a mean of USD 3.34 billion in total assets. Regarding subsidiary banks' risk-taking measures, the mean logarithm of the Z-score is 3.65, the average loan and lease yield is 4.82%, the provision ratio is 0.32%, and the mean leverage ratio is 7.29.

3 Empirical approach and results

The central premise of our paper is the impact of parent debt ratio on the risk-taking of subsidiary banks. In this section, we present evidence that establishes the negative link between the parent debt ratio and subsidiary bank's risk-taking.

For each subsidiary bank, we estimate the following model:

$$RiskTaking_{ijt} = \alpha + \beta ParentDebtRatio_{j,t-4} + \\ + \gamma Controls_{i,t-4} + \delta Controls_{j,t-4} + \eta_t + \epsilon_{ijt} \quad (1)$$

where, for each subsidiary bank i affiliated with BHC j and in quarter t, $ParentDebtRatio$ is calculated as the parent's liability divided by the BHC's total assets, $Controls$ represents a set of subsidiary bank-level and BHC-level control variables described in Section 2.4. The BHC-level control variables are $HQLAratio$ and $TobinQ$. Since most subsidiary banks are not public, we could not calculate the bank-level Tobin Q. Therefore, we use BHC-level Tobin Q as an alternative in our regression model. All explanatory variables are lagged by four quarters to mitigate concerns over look-ahead bias. $RiskTaking$ is a measure of bank's risk-taking originating in quarter t described in Section 2.3. We include subsidiary bank and quarter fixed effects to mitigate omitted-variable bias. In all models, standard errors are clustered at the subsidiary bank level.

Table 3 presents the empirical evidence showing a negative relationship between the parent debt ratio and the subsidiary bank's risk-taking during the 2011–2021Q2 period. Specifically, for a one standard deviation increase in the parent debt ratio (0.033), we observe an expected 9.80% increase in Z-Score (2.97 x 0.033 x 100% = 9.80%), a 0.20% decrease in loan yield (-0.062 x 0.033 x 100% = -0.20%), a 0.16% decrease in provision ratio (-0.048 x 0.033 x 100% = -0.16%), and a 55.08% decrease in leverage (-16.91 x 0.033 x 100% = -55.80%).

To compare magnitudes across outcomes, we scale each of these changes by the standard deviation of the respective outcome variable. The effects are economically meaningful. For the Z-score, whose standard deviation is 0.579, a one-standard-deviation

increase in the parent debt ratio (0.033) is associated with an increase of 0.17 standard deviations (2.97 x 0.033 / 0.579 = 0.17). By the same measure, a one-standard-deviation increase in the parent debt ratio is associated with decreases of 0.13 standard deviations in the loan yield (-0.062 x 0.033 / 0.0157 = 0.13), 0.23 standard deviations in the provision ratio (-0.048 x 0.033 / 0.0068 = -0.23), and 0.18 standard deviations in the leverage ratio (-16.91 x 0.033 / 3.0585 = -0.18).

Taken together, these magnitudes indicate that parent-level debt meaningfully reduces subsidiary banks' risk-taking.

4 Institutional Background and Identification

Identifying how a BHC's parent debt affects its subsidiary banks' risk-taking is difficult because the identification of exogenous changes in BHC's debt structure is an empirical challenge. We address this challenge through the introduction of the SPOE resolution strategy and the gradual introduction of TLAC. requirement in the United States. The episode is useful for two reasons. First, for the BHCs within their scope, SPOE and TLAC changed how limited liability allocates losses between the creditors of the parent and the creditors of its subsidiaries. Second, which BHCs would fall within that scope was not settled in advance. The Dodd–Frank Act did not limit the new regime to the largest banks, and BHCs with total consolidated assets of $50 billion or more, the threshold for resolution planning and enhanced prudential standards, had reason to expect that it would apply to them. When regulators confined TLAC to the eight U.S. global systemically important banks (GSIBs) in December 2015, large non-GSIB BHCs experienced an unexpected relaxation of anticipated regulation. This group, which

studies of resolution reform centered on GSIBs largely overlook, is the focus of our identification strategy.

4.1 Regulatory Background

Before these reforms, the parent and its subsidiaries were resolved as separate legal entities protected by limited liability. A failing bank subsidiary is placed in FDIC receivership on its own; its losses fall on its own creditors and on its shareholder, the parent, and the parent's creditors lose only to the extent that the parent's equity in the subsidiary loses value. Debt issued by the parent and debt issued by the subsidiary bank are therefore exposed to the losses of different legal entities.

SPOE changes this allocation. Under the strategy that the FDIC described in December 2013, only the top-tier holding company enters receivership. Its assets, consisting mainly of its equity in and loans to its subsidiaries, are transferred to a bridge company, whereas its shareholders and unsecured creditors remain in the receivership and absorb losses. The operating subsidiaries, including the insured depository institution, continue as going concerns and are recapitalized with resources from the parent.[2] Losses incurred anywhere in the group are thus pushed up to the parent's creditors, while the external creditors of the subsidiaries are shielded. Parent-level creditors bear losses that ordinary limited liability would have left with the subsidiary.

SPOE is credible only if the parent holds enough equity and long-term unsecured debt to absorb these losses, and TLAC makes it operable. For each covered firm, the Federal Reserve's TLAC rule requires the top-tier holding company to maintain min-

[2]The FDIC notes that the parent's debt "would be used to absorb losses, recapitalize the operating subsidiaries, and allow establishment of NewCo" [Federal Deposit Insurance Corporation, 2013, p. 76617].

imum amounts of external TLAC and of long-term debt issued by the parent.[3] Its "clean holding company" provisions prohibit the parent from issuing short-term debt to third parties, from entering into qualified financial contracts with third parties, and from having liabilities guaranteed by its subsidiaries. For affected BHCs, parent-level debt therefore changed from a financing choice into a regulatory requirement, and the allocation of external debt between the parent and the subsidiaries was no longer determined by the BHC alone. The following table summarizes the regulatory steps; the rest of this section discusses our identification strategy.

4.2 Identification Strategy

The central identification challenge is to isolate exogenous changes in BHCs' parent-level debt. We address this challenge by exploiting the introduction of SPOE and the gradual implementation of TLAC as a natural experiment, treating 2014 and 2015 as a transition period during which uncertainty about the scope of the new regulatory regime was progressively reduced. Three institutional features motivate our identification strategy.

First, SPOE was not designed exclusively for GSIBs, and the Dodd–Frank Act did not settle which firms would have to hold loss-absorbing debt. The Orderly Liquidation Authority can be applied to any financial company whose failure would threaten U.S. financial stability, and all BHCs with total consolidated assets of $50 billion or more were already required to file resolution plans under Section 165(d). Section 165 also directs the Federal Reserve to apply enhanced prudential standards to the same set of BHCs

[3] External TLAC must equal at least the greater of 18 percent of risk-weighted assets (RWA), plus a buffer of 2.5 percent and the firm's GSIB surcharge, and 7.5 percent of total leverage exposure, plus a buffer of 2 percent. Eligible external long-term debt must equal at least the greater of 6 percent of RWA plus the GSIB surcharge and 4.5 percent of total leverage exposure. Eligible debt must be unsecured, "plain vanilla," governed by U.S. law, and have a remaining maturity of at least one year [Board of Governors of the Federal Reserve System, 2017].

Table 1: Timeline of the SPOE and TLAC reforms

Date	Event
2008–2011	G20 calls for effective resolution regimes and creates the FSB; the FSB's *Key Attributes* establish bail-in as a core resolution power [G20, 2008, 2009a,b, Financial Stability Board, 2011, G20, 2011].
July 2010	Dodd–Frank Act: Section 165(d) requires resolution plans from BHCs with $50 billion or more in assets; Title II creates the Orderly Liquidation Authority.
Dec. 2012	FDIC and Bank of England joint paper on resolving GSIFIs from the top [Bank of England and Federal Deposit Insurance Corporation, 2012].
Sept. 2013	G20 asks the FSB to propose loss-absorbing capacity standards for global systemically important institutions [G20, 2013].
Dec. 18, 2013	FDIC notice on the SPOE strategy; does not specify which firms must hold parent-level loss-absorbing debt [Federal Deposit Insurance Corporation, 2013].
Feb. 18, 2014	Enhanced prudential standards (EPS) final rule for BHCs with $50 billion or more; no long-term debt requirement [Board of Governors of the Federal Reserve System, 2014].
Nov. 10, 2014	FSB consultative TLAC term sheet, limited to GSIBs [Financial Stability Board, 2014].
July 20, 2015	GSIB surcharge rule identifies eight U.S. GSIBs [Board of Governors of the Federal Reserve System, 2015].
Oct. 30, 2015	Federal Reserve proposes TLAC and long-term debt requirements for U.S. GSIBs only [Board of Governors of the Federal Reserve System, 2016].
Nov. 2015	FSB final TLAC standard; G20 endorsement at Antalya [Financial Stability Board, 2015, G20, 2015].
Dec. 15, 2016	Final TLAC rule; compliance from January 1, 2019 [Board of Governors of the Federal Reserve System, 2017].
May 2018	EGRRCPA raises the general EPS threshold to $250 billion (after our sample).

and allows it to tailor those standards across firms; the TLAC rule was later adopted under this authority. Because SPOE imposes losses on the parent while preserving the operating subsidiaries, it can succeed only if the parent holds sufficient loss-absorbing debt and equity. A parent-level debt requirement could therefore plausibly have been imposed on every BHC above $50 billion, and the FDIC's notice of SPOE in December 2013 did not say otherwise. When the notice was published, it was unclear whether large non-GSIB BHCs would also be expected to prepare for SPOE and to issue additional long-term debt at the parent level.

Second, the EPS rule of February 2014, to some extent, reduced this uncertainty. The rule was the Federal Reserve's main implementation of Section 165 for all BHCs with total consolidated assets of $50 billion or more, and it subjected GSIBs and large non-GSIB BHCs to a common set of capital, liquidity, and risk-management standards. Yet it imposed no requirement related to SPOE, TLAC, or long-term debt. Had the Federal Reserve intended to require loss-absorbing debt from every BHC above $50 billion, the EPS rule would have been a natural vehicle for doing so; its silence on this point was a first signal that such a requirement might be confined to a narrower set of firms. The rule did not, however, eliminate the uncertainty. Section 165 still allowed the Federal Reserve to add standards for any covered company, and the international work on loss-absorbing capacity that the G20 had requested was not yet complete. Whether large non-GSIB BHCs would have to hold loss-absorbing debt at the parent therefore remained an open question after February 2014.

The third feature is that the sequential design of the TLAC framework resolved this uncertainty during 2014Q4 and 2015 by ultimately excluding large non-GSIB BHCs from the scope of both SPOE and TLAC. Because equity and long-term unsecured debt at the parent are necessary for SPOE to operate, the scope of TLAC effectively determined

the scope of SPOE. The FSB's consultative term sheet of November 2014 limited the international TLAC standard to GSIBs, although national authorities remained free to extend similar requirements to other firms. The Federal Reserve's identification of the eight U.S. GSIBs in July 2015 and its proposal of October 30, 2015, which applied the TLAC and long-term debt requirements to those eight firms only, then confirmed that large non-GSIB BHCs would be exempt from the U.S. TLAC requirement; the FSB finalized the international standard in November 2015. Large non-GSIB BHCs thus learned that they would not have to hold parent-level loss-absorbing debt, while they remained subject to the same enhanced prudential standards as GSIBs. For these BHCs, the announcement removed an expected regulatory reason to issue debt at the parent without changing the other rules that apply to them. It is this change in expectations that we exploit.

Therefore, we classify BHCs into three groups based on total consolidated assets as of 2014Q1. The treatment group comprises large non-GSIB BHCs with total consolidated assets of $50 billion or more (TreatBHCs), which were expected to be covered but were exempted. The first comparison group comprises BHCs with total consolidated assets below $50 billion, which were never expected to be covered. The second comparison group comprises the eight U.S. GSIBs, which were expected to be covered and remained so. Once the narrower scope became clear, parent-level debt no longer served a regulatory purpose for large non-GSIB BHCs in the treatment group, which should therefore reduce their reliance on it relative to *both* comparison groups.

GSIBs also serve as a placebo treatment group because they are more comparable to large non-GSIB BHCs and are therefore more likely to share similar time-varying trends. Large non-GSIB BHCs differ from smaller BHCs in terms of size, complexity, and supervisory oversight, so a comparison with smaller BHCs alone could capture time-

varying shocks specific to large BHCs rather than the effects of SPOE and TLAC. GSIBs share these characteristics and are subject to the same EPS regime as large non-GSIB BHCs, making them more likely to experience similar time-varying shocks. The key time-varying institutional changes that the two groups were expected to share, but that distinguish them from smaller BHCs, are the introduction of SPOE and TLAC. If our results were driven by trends or shocks common to large BHCs rather than by SPOE and TLAC, a comparison between GSIBs and smaller BHCs should yield similar estimates.

Figure 4 and Figure 5 provide evidence consistent with this design. Figure 5 shows that the parent debt ratios of the treatment and control groups move in parallel until 2014, when the SPOE strategy and EPS rule took effect. After 2014, the ratio for the treatment group declines sharply, whereas the ratio for the control group remains stable, with a slight upward trend. Specifically, the parent debt ratio of the treatment group decreases from 3.5% in 2014Q1 to 3.0% in 2015Q4, whereas that of the control group increases from 0.4% to 0.5%. Figure 5 shows that the parent debt ratio of GSIBs rises sharply after the U.S. TLAC rule was proposed at the end of 2015, from 9.0% in 2015Q4 to 9.6% in 2017Q3.

We first examine whether risk-taking increases more for large non-GSIB BHCs than for BHCs in the control group after the proposal of the U.S. TLAC rule.

4.3 Econometric framework: Difference-in-differences estimator

We estimate a difference-in-differences model using 2012–2013 as the pre-period and 2016–2017 as the post-period. For each BHC, we specify our baseline model as follows:

$$RiskTaking_{ijt} = \alpha + \beta\ Post_t + \gamma\ TreatBHC_{ij} \times Post_t + \lambda\ GSIBs_{ij} \times Post_t + \delta\ Controls_{ijt} + \epsilon_{ijt} \quad (2)$$

where *Post* is a dummy variable equal to 1 for the 2016–2017 post-period and 0 for the 2012–2013 pre-period, and $TreatBHC$ is a dummy variable equal to 1 for large non-GSIB BHCs subject to the Enhanced Prudential Standards but not SPOE strategy or TLAC requirement. $GSIBs$ is an indicator equal to 1 for GSIBs. $Controls$ is a set of subsidiary bank- and BHC-level control variables. α_i denotes subsidiary bank fixed effects. The model does not include the stand-alone $TreatBHC$ dummy variable, because it is subsumed by the subsidiary bank-specific fixed effect α_i. $RiskTaking$ denotes one of our measures of subsidiary bank risk-taking: the loan yield, the provision ratio, subsidiary bank leverage, or the Z-score.

We exclude 2014–2015 from the estimation sample, treating it as the transition period during which the scope of SPOE and TLAC was resolved and parent-level debt structures were adjusted. This follows common practice in difference-in-differences designs with a gradually revealed shock, in which observations recorded while the shock is still unfolding are excluded to avoid attenuating the estimated treatment effect.

We include the interaction $GSIBs \times Post$ alongside our coefficient of interest, $TreatBHC \times Post$, for the following reason. It serves as a placebo test of our identifying assumption. GSIBs were subject to SPOE and EPS from the outset, and their TLAC obligation never depended on how the scope of the rule was resolved during the drafting period. If our identification is valid, the subsidiary banks of GSIBs should exhibit no discernible shift in risk-taking over the same window, and λ should be statistically indistinguishable

from zero. A significant λ would instead suggest that a confounding factor common to all large BHCs, rather than the resolution of TLAC uncertainty specific to non-GSIB BHCs, drives our results.

To address the downward bias in standard errors that arises in difference-in-differences estimation with serially correlated outcomes, we follow Bertrand et al. [2004] and take the averages of the observations in our sample in the pre-period (2012–2013) and in the post-period (2016–2017). We will conduct a range of robustness checks and falsification tests, including alternative sample periods, to verify our results and rule out alternative explanations. In one such test, we use the transition period (2014–2015) as the post-period to illustrate the gradual decline in the parent debt ratio shown in Figure 3.

Table 2, Panel B, summarizes the sample descriptive statistics of our key variables for the 2012–2013 and 2016–2017 periods. An average subsidiary bank in our natural experiment sample has log total assets of 14.78 and a parent debt ratio of 2.0%. The average Tobin's Q for BHCs is 1.03. The mean value of HQLA ratio is 15.10%. Regarding subsidiary banks' risk-taking measures, the mean logarithm of the Z-score is 3.68, the average loan and lease yield is 4.92%, the provision ratio is 0.36%, and the mean leverage ratio is 7.35. Table 2, Panel C summarizes the collapsed pre- and post-period sample of the treatment group, and Table 5 reports the regression results. Across all risk-taking measures, relative to BHCs below the $50 billion threshold, $50 billion-plus non-GSIB BHCs ($TreatBHC \times Post$) increase risk-taking following the resolution of TLAC uncertainty, over and above the effect of the preceding EPS shock, relative to BHCs below the $50 billion threshold. In contrast, GSIBs ($GSIB \times Post$) exhibit no significant change, which is precisely the pattern predicted by our identification strategy. The coefficient for the treatment effect ($TreatBHC \times Post$) is statistically significant at the 1% level. Model (1) of Table 4 shows that, on average, large non-GSIB BHCs experienced

an increase in loan yield that was 0.006 greater than that of BHCs in the control group, whose loan yield declined modestly over the same period. This effect is robust to the inclusion of bank-level controls in Model (2). Similarly, relative to the control group, large non-GSIB BHCs experienced an average increase of 0.003 in the provision ratio (Model (4)) and of 0.791 in the leverage ratio (Model (10)). Model (8) further shows that large non-GSIB BHCs experienced a decline in the Z-score that was, on average, 0.236 greater than that of the control group. Collectively, these results indicate that large non-GSIB BHCs increased their risk-taking following the introduction of the TLAC rule.

To compare magnitudes across outcomes, we scale each treatment effect by the standard deviation of the respective outcome variable. The effects are economically meaningful. For the Z-score, whose standard deviation is 0.808, the treatment effect corresponds to a decrease of 0.29 standard deviations (-0.236/0.8080 = -0.29). By the same measure, the treatment effect is associated with increases of 0.18 standard deviations in the loan yield (0.0060/0.0333=0.18), 0.29 standard deviations in the provision ratio (0.0030/0.0104=0.29), and 0.24 standard deviations in the leverage ratio (0.7910/3.3340=0.24).

These magnitudes are comparable to those from the baseline panel regressions, in which a one-standard-deviation increase in the parent debt ratio is associated with changes of 0.13 to 0.23 standard deviations across the same outcomes. In the difference-in-differences analysis, the treatment effects range from 0.18 to 0.29 standard deviations, with signs opposite to the panel estimates, as expected given the decline in parent debt among BHCs in the treatment group. The similarity between the two sets of estimates suggests that the panel associations are not driven primarily by the endogeneity of parent debt.

Taken together, the evidence is consistent with a causal effect of parent debt on the risk-taking of subsidiary banks. As TLAC uncertainty was resolved in their favor, BHCs in the treatment group reduced their parent debt and their subsidiaries increased risk-taking. GSIBs, for which the resolution of TLAC scope carried no new information, exhibit no comparable change.

5 Model Setup

We consider a single-period model of a *bank holding company* (BHC) consisting of three entities: a parent company, its subsidiary bank, and its non-bank subsidiary. The parent company wholly owns and controls the two subsidiaries, and the risk-neutral shareholder of the parent company makes decisions.

Parent company

The parent company's capital structure consists of outside equity $E_p > 0$ and external debt $D_p \geq 0$ raised at the parent level. The parent's total funding is $E_p + D_p$. While the BHC will also raise external debt with its subsidiaries, e.g., deposits, all outside equity in the BHC resides at the parent level. For simplicity, we assume that the parent company's debt is priced at the risk-free rate r_f, which can be due to an implicit too-big-to-fail guarantee at the holding company level. As will be shown below, the solvency of the parent company is also featured in the equilibrium of the model.

The parent allocates a fraction θ of its total funding to the bank as equity, and the remainder to the non-bank subsidiary. The allocation parameter θ captures the relative profitability and the sizes of the subsidiary bank and the other subsidiary.[4]

[4]It can be shown that the main results of the theoretical model is not affected by the value of θ.

Subsidiary bank

The subsidiary bank issues retail deposits D_b, over which the subsidiary bank pays a gross deposit rate, r_d. The bank subsidiary receives equity financing E_b from the parent. As the parent company allocates θ of its funding to the subsidiary bank, we have $E_b = \theta(E_p + D_p)$ The subsidiary bank's balance sheet is normalized to unity: $E_b + D_b = 1$. The bank has no minority shareholders: all equity is injected by the parent company. Bank deposits carry a convenience yield $v \in [0, r_f]$ (e.g., transactions services, liquidity), which is the social value of bank deposit-taking. Depositors therefore accept a below-market rate:

$$r_d = r_f - v. \tag{1}$$

For the fixed size of subsidiary bank's balance sheet, deposits and parent company debt are connected as follows:

$$D_b = 1 - \theta\,(E_p + D_p), \qquad \frac{dD_b}{dD_p} = -\,\theta. \tag{2}$$

That is, each additional unit of parent debt replaces θ units of bank deposits.

The subsidiary bank invests its unit balance sheet in a portfolio whose risk-return profile is governed by a single choice variable $R > 1$, interpreted as the gross portfolio return in the "good" state. The portfolio has binary outcomes:

$$\widetilde{R}_b \;=\; \begin{cases} R & \text{with probability } p(R), \\ 0 & \text{with probability } 1 - p(R). \end{cases} \tag{3}$$

The success probability is strictly decreasing in R, capturing the fundamental risk-return trade-off. The model will focus on the risk choice of the subsidiary bank side, because

data are more readily observable to facilitate the test of the theory. For simplicity, we assume that the success probability takes a linear form

$$p(R) = 1 - \frac{\beta}{2} R, \tag{4}$$

where $\beta \in (0, 2/R]$ so that $p(R) \in [0, 1]$. The expected portfolio return is therefore

$$\mathbb{E}[\widetilde{R}_b] = R \left(1 - \frac{\beta}{2} R\right) = R - \frac{\beta}{2} R^2, \tag{5}$$

which is maximized at $R^* = \frac{1}{\beta}$, yielding $\mathbb{E}[\widetilde{R}_b^*] = \frac{1}{2\beta}$. We require $R > (r_f - v)\, D_b$ for the bank to be solvent in the good state, i.e., the portfolio must generate enough to cover depositors when it succeeds.

Non-bank subsidiary

The parent company also wholly owns another subsidiary, which can be a non-bank financial subsidiary.[5] The non-bank subsidiary operates a *constant-returns-to-scale* (CRS) technology: each unit of its investment yields a gross return

$$\widetilde{X} = \begin{cases} X & \text{if the bank portfolio returns } R, \\ X - \Delta & \text{if the bank portfolio returns } 0, \end{cases} \tag{6}$$

where $X > 0$ is the *per-unit* gross return on non-bank investment and $\Delta \in [0, X]$ parameterizes the *intra-BHC correlation*. When $\Delta = 0$, the non-bank's return is independent of the bank's performance; when $\Delta = X$, the non-bank is worthless precisely when the bank fails. For the baseline analysis, we will start with a special case of independent

[5] This subsidiary can be a non-bank, such as an insurance arm or broker-dealer, or a different banking subsidiary. We will call this additional subsidiary the non-bank subsidiary to fix the idea.

returns and $\Delta = 0$, which in turn requires $X = r_f$. If $X > r_f$, the parent could raise debt at r_f and invest it in the non-bank at X, an arbitrage that would result in unbounded parent leverage. The analysis isolates the placement of external debt rather than the profitability of non-bank expansion. For simplicity, we assume this subsidiary to be fully equity financed so that we focus on the allocation of external debt between the parent company and the banking subsidiary.[6]

Shareholder's risk-taking decision

The parent company's shareholder chooses the bank's portfolio risk R to maximize their expected equity value. Specifically, the parent company's asset value is the sum of the residual cash flows it receives from both subsidiaries. From the subsidiary bank, the parent receives the residual after depositors are repaid,

$$V_b \;=\; \max\{\widetilde{R}_b - r_d\, D_b,\; 0\}. \tag{7}$$

Combined with the non-bank subsidiary value,

$$V_{nb} \;=\; \tilde{X}(1-\theta)(E_p + D_p) = r_f(1-\theta)(E_p + D_p). \tag{8}$$

the parent's total assets are $V_b + V_{nb}$ and the equity holders receive the residual after the parent-level debt is repaid:

$$\Pi = \max\{V_b + V_{nb} - r_f\, D_p,\; 0\}. \tag{9}$$

[6]Given the risk-free investment of the non-bank subsidiary, adding leverage to this subsidiary would not change its firm value.

5.1 First-best Allocation

We start our analysis by establishing a first-best benchmark where a benevolent social planner has direct control over *both* the bank risk level *and* the BHC's capital structure. In particular, the social planner chooses R (and thereby the bank's asset risk) as well as the amount of parent-level debt (and thereby the amount of retail deposit issued) to maximize the NPV of bank assets and the social value of deposits simultaneously:[7]

$$W(R,\, D_b) \;=\; R\,(1 - \frac{\beta}{2}R) + v\,D_b, \tag{10}$$

where the first term captures the expected bank asset value and the second term captures the total convenience yield generated by deposit-taking. The first-order condition in R yields the efficient risk level

$$R^{FB} = \frac{1}{\beta}. \tag{11}$$

Since $\partial W/\partial D_b = v > 0$, the first-best also calls for $D_b = 1$ to maximize deposit funding. As the social planner exerts direct control over both the risk level and the capital structure, the two optimal choice of the two are disjoint and summarized as below.

$$\left(R^{\text{FB}},\, D_b^{\text{FB}}\right) = \left(\frac{1}{\beta},\, 1\right), \qquad W^{\text{FB}} = \frac{1}{2\beta} + v. \tag{12}$$

5.2 Constrained Social Optimum

We now consider the case where the planner can dictate the BHC's capital structure—in particular, the level of parent debt D_p—but has no direct control over the bank's risk level R, which is chosen by the parent company's shareholders. That is, R is not

[7] Note that the non-bank investment is zero-NPV and therefore does not enter the welfare function.

contractible, but the capital structure is. The social planner can only indirectly affect the shareholder's risk choice by varying the bank holding company's capital structure.

This creates a tension: the planner would like to maximize deposits (to capture the convenience yield) but must respect the shareholders' privately optimal risk-taking response to the resulting capital structure. In this sense, the social planner is constrained by the shareholders' incentive compatibility.

We frame the analysis in terms of the parent's debt level D_p, which is the planner's instrument. Each dollar of parent debt crowds out θ dollars of deposits, so the planner's constrained maximization problem is

$$\begin{aligned} \max_{D_p} \quad & W\big(R^{\text{priv}}(D_p),\ D_b(D_p)\big) \\ \text{s.t.} \quad & R = R^{\text{priv}}(D_p) \text{ is the shareholders' best response,} \end{aligned}$$

where $D_b(D_p) = 1 - \theta\,(E_p + D_p)$ from (2) and $R^{\text{priv}}(D_p)$ is derived below.

5.2.1 Shareholder's private risk choice

We maintain the regularity condition that the bank is solvent in the good state: $R > (r_f - v)\,D_b$.

- **Good state** (prob. $1 - \frac{\beta}{2}R$). The bank repays depositors and remits the surplus $R - (r_f - v)\,D_b$ to the parent, and the parent's total assets equal $R - (r_f - v)\,D_b + r_f\,(1-\theta)\,P$.
- **Bad state** (prob. $\frac{\beta}{2}R$). The bank is insolvent, and the parent's only asset is its equity value in the non-bank subsidiary, $r_f\,(1-\theta)\,P$.

The parent defaults in the bad state if and only if $r_f\,(1-\theta)\,P < r_f\,D_p$, i.e., using $P = E_p + D_p$,

$$D_p > \bar{D}_p \;\equiv\; \frac{(1-\theta)\,E_p}{\theta}. \tag{13}$$

Since the non-bank earns exactly the risk-free rate, the parent is solvent in the bad state when its debt does not exceed the resources parked in the non-bank subsidiary, of which a fraction $(1-\theta)$ of every parent dollar accrues.

We use backward induction and derive the shareholder's private choice of risk for a given parent-level debt D_p. The cutoff $\bar{D}_p$ creates two cases.

Case (1): Parent always solvent ($D_p \le \bar{D}_p$)

When there is limited amount of parent-level debt and the parent company never defaults, equity holders receive a positive payoff in both states. The expected equity value of the whole BHC is as follows:

$$\mathbb{E}[\widetilde{\pi}_E] = \big(1 - \tfrac{\beta}{2}R\big)\big[R - (r_f - v)\,D_b\big] + r_f\big[(1-\theta)E_p - \theta\,D_p\big]$$

The first-order condition with respect to R gives

$$R_1^{\text{priv}}(D_p) = \frac{1}{\beta} + \frac{(r_f - v)\,D_b}{2} > R^{FB} \tag{14}$$

The risk level exceeds the efficient level because of the bank-level limited-liability distortion. For $D_b = 1 - \theta\,(E_p + D_p)$, we have

$$\frac{\partial R_1^{\text{priv}}(D_p)}{\partial D_p} = -\,\frac{\theta\,(r_f - v)}{2} < 0. \tag{15}$$

That is, when the parent is solvent, increasing D_p reduces the risk chosen by shareholders: each extra dollar of parent debt crowds out θ dollars of deposits (D_b), mitigating the bank-level limited-liability distortion in proportion to the pass-through θ. Meanwhile, parent debt itself creates no risk-shifting incentive because the parent always repays.

Note also that D_p does not appear directly in (14); it enters only through $D_b = 1 - \theta\,(E_p + D_p)$. The parent company's debt obligation is state-independent when the parent is solvent, so it does not distort the shareholders' marginal risk–return trade-off.

Case (2): Parent defaults in the bad state ($D_p > \bar{D}_p$)

When the parent is insolvent in the bad state, equity is worthless in that state, and

$$\mathbb{E}[\widetilde{\pi}_E] = \big(1 - \tfrac{\beta}{2}R\big)\big[R - (r_f - v)\,D_b + r_f(1-\theta)(E_p + D_p) - r_f D_p\big]. \tag{16}$$

The first-order condition yields

$$R_2^{\text{priv}}(D_p) = \frac{1}{\beta} + \frac{(r_f - v)\,D_b + r_f\big(\theta\,D_p - (1-\theta)E_p\big)}{2} > R_1^{\text{priv}}(D_p) > R^{FB}. \tag{17}$$

Note that $\theta D_p - (1-\theta)E_p = \theta\,(D_p - \bar{D}_p) > 0$ in the default region, so that $R_2^{\text{priv}}(D_p) > R_1^{\text{priv}}(D_p)$ — the parent-level term adds to the distortion, Further note that $R_1^{\text{priv}}(D_p) = R_2^{\text{priv}}(D_p)$ for $D_p = \bar{D}_p$, and therefore the privately optimal risk level is continuous in D_p.

When the parent can default, increasing D_p increases the risk chosen by shareholders.

$$\frac{\partial R_2^{\text{priv}}(D_p)}{\partial D_p} = \frac{\theta\,v}{2} > 0. \tag{18}$$

Intuitively, an additional unit of parent debt adds r_f to the face-value burden borne in the good state, removes $\theta\,(r_f - v)$ of deposit obligations, and contributes $(1-\theta)\,r_f$ of additional non-bank collateral. The net effect is $r_f - \theta(r_f - v) - (1-\theta)r_f = \theta\,v > 0$: per unit of deposits displaced, replacing deposits with parent debt raises the BHC's total face-value burden by exactly the convenience yield v—the "subsidy" embedded in deposit funding.

Combining Cases (1) and (2) reveals that the marginal impact of parent-level debt on bank risk-taking flips the sign at $D_p = \bar{D}_p$, and the risk is minimized for $D_p = \bar{D}_p$. Proposition 5.2.1 summarizes the results.

The shareholders' privately optimal risk level is

$$R^{\text{priv}}(D_p) = R^* + \frac{(r_f - v)\,D_b + r_f\,\theta\,\max\{D_p - \bar{D}_p,\ 0\}}{2}, \tag{19}$$

which is continuous and piecewise linear in D_p, with

$$\frac{\partial R^{\text{priv}}}{\partial D_p} = \begin{cases} -\dfrac{\theta\,(r_f - v)}{2} < 0 & \text{if } D_p \le \bar{D}_p \ \text{ (parent solvent)}, \\[2ex] \dfrac{\theta\,v}{2} > 0 & \text{if } D_p > \bar{D}_p \ \text{ (parent insolvent)}. \end{cases} \tag{20}$$

Risk is minimized at the solvency threshold $D_p = \bar{D}_p$.

5.2.2 Constrained social optimum of parent-level debt D_p

Having solved the shareholder's choice of risk for a given parent-level debt D_p, we now solve the constrained social optimum.

First note that when $D_p > \bar{D}_p$, the social welfare strictly decreases in D_p. A greater amount of parent debt has two impacts in this case: first, it partly substitute retail deposits as the subsidiary bank's funding, and therefore reduce the social value of convenience yield. Second, as established in the last subsection, the parent debt in this case also increases excessive risk-taking. More parent-level debt simultaneously increases the distortion in risk-taking decisions and destroys deposit value. Therefore, we know that in a constrained optimum, it must hold that $D_p \leq \bar{D}_p$ so that the parent company does not default.

At the constrained social optimum, the parent-level debt must satisfy $D_p \leq \bar{D}_p$ so that the parent company does not default.

We now turn to the case where $D_p \leq \bar{D}_p$. A trade-off arises in this case where the parent company does not default: on the one hand, more parent-level debt partly replaces retail deposits and decreases the social value of convenience yield. On the other hand, parent-level debt allows more risk-taking losses at the subsidiary level to be passed onto the shareholder of the parent company, which mitigates the limited liability distortion and reduces risk-taking. The optimal parent-level debt strikes a balance between the two.

The welfare consists of expected value of the bank's loan portfolio and the total convenience yield created by deposits $W(D_p) = R\left(1 - \frac{\beta}{2}R\right) + v\, D_b$, which when evaluated at $R = R_1^{Priv}$ gives

$$W(D_p|R = R_1^{Priv}) = \frac{1}{2\beta} - \frac{\beta\,(r_f - v)^2}{8}\, D_b^2 + v\, D_b. \tag{21}$$

Differentiating with respect to D_p and using $dD_b/dD_p = -\theta$, we obtain

$$\frac{dW}{dD_p} = \theta \left[\frac{\beta (r_f - v)^2}{4} D_b \; - \; v \right], \tag{22}$$

with the second order derivative confirming strict concavity.

$$\frac{d^2W}{dD_p^2} = - \frac{\theta^2 \beta (r_f - v)^2}{4} < 0.$$

The first-order derivative reveals the trade-off between the marginal benefit of reducing risk distortion (the first term in the bracket) against the marginal cost of losing deposits (second term in the bracket). The first order condition gives an unconstrained maximizer

$$D_p^\circ = \frac{1}{\theta} \left[1 - \frac{4v}{\beta (r_f - v)^2} \right] - E_p. \tag{23}$$

There are also two possible corner solutions, $D_p^* = 0$ and $D_p^* = \bar{D}_p$, depending on the parameters of the model. Specifically, if $D_p^\circ < 0$, the welfare is decreasing in D_p for the entire interval of $[0, \bar{D}_p]$, so the optimal parent-level debt is zero. That is, the convenience yield that deposits carry is high and/or the risk-shifting at the subsidiary bank level is weak, so that the parent-level debt is set to zero to maximize socially valuable deposit creation. If $D_p^\circ > \bar{D}_p$, the optimal parent-level debt equals $\bar{D}_p$. In this opposite case, the planner would prefer even more parent debt to further reduce subsidiary-level risk-shifting, but setting $D_p > \bar{D}_p$ would lead to limited liability distortion at the parent level and undermines the purpose of reducing risk-taking. Proposition 5.2.2 summarizes these results.

The constrained-optimal level of parent debt trades off between the risk-reduction at the subsidiary level and the social value of deposit creation and is as follows:

$$D_p^* = \begin{cases} 0 & \text{if } D_p^\circ < 0, \\ D_p^\circ = \frac{1}{\theta}\left[1 - \frac{4v}{\beta\,(r_f - v)^2}\right] - E_p & \text{if } 0 \le D_p^\circ \le \bar{D}_p, \\ \bar{D}_p = \frac{(1-\theta)\,E_p}{\theta} & \text{if } D_p^\circ > \bar{D}_p. \end{cases} \tag{24}$$

At the constrained social optimum, parent-level debt can be positive to limit bank-level leverage but is capped at $\bar{D}_p$ to avoid double-layered limited liability distortion, i.e., both at the subsidiary bank and parent levels. The exact amount of parent-level debt balances risk reduction against the social value of convenience yield carried by deposits. At the interior solution $D_p^* = D_p^\circ$, the risk level is $R^{FB} + \frac{2v}{\beta\,(r_f - v)}$, which exceeds R^{FB}: the planner tolerates some excessive risk-taking to preserve the deposit franchise value. The corresponding welfare level is

$$W^* = \frac{1}{2\beta} + \frac{2v^2}{\beta\,(r_f - v)^2},$$

and the welfare loss relative to first-best is

$$W^{\mathrm{FB}} - W^* = v - \frac{2v^2}{\beta\,(r_f - v)^2}.$$

The reduced welfare reflects the shortfall in deposits ($D_b < 1$ in the constrained optimum) as well as the increased risk-taking incentives.

5.3 Comparative statics and further testable hypotheses

The comparative statics of the interior optimum with respect to the convenience yield v and the relative size of the banking subsidiary θ reveal relationships that potentially empirically testable.

At the interior optimum, the parent-level debt decreases in the convenience yield v, whereas bank risk increases in v:

$$\frac{\partial D_p^\circ}{\partial v} < 0, \qquad \frac{\partial R^\circ}{\partial v} > 0. \tag{25}$$

The parent-level debt decreases in the relative size of the banking subsidiary θ, whereas bank risk is independent of θ:

$$\frac{\partial D_p^\circ}{\partial \theta} < 0, \qquad \frac{\partial R^\circ}{\partial \theta} = 0. \tag{26}$$

A higher convenience yield raises the social value of the marginal deposit, so it is desirable to fund the bank with more deposits and less parent-level debt. Cheaper deposits directly lower the bank's repayment burden, but the induced expansion of deposit funding aggravates the distortion of limited liability, so equilibrium risk rises. A bigger non-bank subsidiary entails more parent debt to fund its operation. This, however, does not change the leverage of the banking subsidiary and therefore does not change the risk level.

6 Conclusion

This paper provides evidence on how parent debt affects subsidiary banks' risk-taking. Using a comprehensive dataset covering all domestic U.S. commercial banks and their corresponding BHCs from 2011 to 2021Q2, combined with a difference-in-differences analysis using the gradual introduction of SPOE and TLAC as a natural experiment, we find that increased parent debt is associated with reduced risk-taking by subsidiary banks. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank's risk profile.

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Table 1: Variable definitions and sources of data used in the study. This table summarizes the variable definitions and sources of data used in our study.

Variables	Definition
Parent Debt Ratio	Ratio of the parent's non-deposit debt liabilities to the BHC's assets, measured quarterly as: (BHCP2309 (parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332 (parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368 (parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062 (parent company's subordinated notes and debentures) + BHCP0279 (parent company's securities sold under agreements to repurchase)) / BHCK2170 (BHC's total assets)
Loan Lease Yield	Ratio of the subsidiary bank's total interest income from loans to its total loans: (RIAD4010 (total interest and fee income on loans)+ RIAD4065) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or (RIAD4010 (total interest and fee income on loans)+ RIAD4065)/ RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041. Because RIAD4010 and RIAD4065 are reported as year-to-date income, we convert these variables into quarterly amounts before calculating the ratio.
Provision Ratio	Ratio of provisions to total loans: RIAD4230 (provisions for credit losses) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or RIAD4230 (provisions for credit losses) / RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041. Because RIAD4230 is reported as year-to-date income, we convert this variable into quarterly amount before calculating the ratio.
Leverage	Leverage is the subsidiary bank's debt-to-equity ratio: RCON2948 (bank subsidiary's total liabilities, FFIEC 041) / (RCON2170 (bank subsidiary's total assets, FFIEC 041) − RCON2948 (bank subsidiary's total liabilities, FFIEC 041)), or RCFD2948 (bank subsidiary's total liabilities, FFIEC 031) / (RCFD2170 (bank subsidiary's total assets, FFIEC 031) − RCFD2948 (bank subsidiary's total liabilities, FFIEC 031))
Z-Score	Z-Score is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as total equity (RCFD3210, total bank equity capital at the end of the current period) divided by total assets (RCFD2170) for FFIEC 031, or total equity (RCON3210, total bank equity capital at the end of the current period) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
Z-Score Narrow	Z-Score Narrow is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as book equity (RCFD3632 (retained earnings) + RCFD3839 (surplus, excluding all surplus related to preferred stock) + RCFD3230 (common stock)) divided by total assets (RCFD2170) for FFIEC 031, or book equity (RCON3632 (retained earnings) + RCON3839 (surplus, excluding all surplus related to preferred stock) + RCON3230 (common stock)) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
Ln TA	Total assets is a proxy for subsidiary bank size. It is the logarithm of the subsidiary bank's total assets: ln(RCON2170 (bank subsidiary's total assets, FFIEC 041)), or ln(RCFD2170 (bank subsidiary's total assets, FFIEC 031))

HQLA Ratio	HQLA_ratio is the ratio of the BHC's high-quality liquid assets to the BHC's total assets: (BHCM3531 (U.S. Treasury securities) + BHCM3532 (U.S. government agency obligations, excluding mortgage-backed securities) + BHCKG379 (residential pass-through MBS issued or guaranteed by FNMA, FHLMC, or GNMA) + BHCKG380 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, including CMOs, REMICs, and stripped MBS) + BHCKK197 (commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies) + BHCKG301 (residential pass-through securities guaranteed by GNMA, held to maturity) + BHCKG303 (residential pass-through securities guaranteed by GNMA, available for sale) + BHCKG305 (residential pass-through securities issued by FNMA and FHLMC, held to maturity) + BHCKG307 (residential pass-through securities issued by FNMA and FHLMC, available for sale) + BHCKKX53 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, held to maturity) + BHCKKX55 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, available for sale) + BHCKG313 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity) + BHCKG315 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale) + BHCKK143 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, held to maturity) + BHCKK145 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, available for sale) + BHCKK151 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity) + BHCKK153 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale) + BHCKHT51 (U.S. government agency and sponsored agency obligations, held to maturity) + BHCKHT53 (U.S. government agency and sponsored agency obligations, available for sale) + RCFD0090 (balances due from Federal Reserve Banks) for FFIEC 031, or RCON0090 (balances due from Federal Reserve Banks) for FFIEC 041 + BHCK0213 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, held to maturity) + BHCK1287 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, available for sale)) / BHCK2170 (BHC's total assets)
TobinQ	Tobin's Q is the group-level consolidated Tobin's Q, defined as the ratio of the market value of the BHC (MV) to the replacement cost of its assets (RC). It is estimated as tobinq_ultp = (AT + ME − BE) / AT, where AT is total assets, ME is the market value of equity at year end (stock price multiplied by common shares outstanding), and BE is book equity (stockholders' equity + deferred taxes + investment tax credit − preferred stock). Calculated using the WRDS code; see link for details: https://wrds-www.wharton.upenn.edu/pages/wrds-research/applications/risk-and-valuation-measures/tobins-q-altman-z-score-and-companys-age/

Table 2

Summary statistics of key characteristics. This table summarizes the sample statistics of key subsidiary bank-level and BHC-level characteristics used in our study. Panel A uses the full sample of subsidiary banks and BHCs during the 2011-2021Q2 period. Panel B uses the reduced sample for difference-indifferences models over the 2012-2013 (pre-deregulation) and 2016-2017 (post-deregulation) periods. Panel C uses the collapsed sample of BHCs in the treatment group for difference-indifferences models over the 2012-2013 (pre-deregulation) and 2016-2017 (post-deregulation) periods. The full sample comprise 298 unique BHCs and 579 subsidiary banks.

Panel A: Full Sample (2011-2021Q2)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	N	mean	sd	p10	p25	p50	p75	p90	min	max
Loan Lease Yield	8,907	0.0482	0.0157	0.0372	0.0415	0.0459	0.0509	0.0571	0.0184	0.1364
Provision Ratio	8,907	0.0032	0.0068	-0.0011	0.0002	0.0016	0.0037	0.0087	-0.0099	0.0399
Z-Score	9,871	3.6525	0.5889	3.0928	3.3643	3.6045	3.8726	4.2256	2.0747	6.2843
Z-Score Narrow	9,871	3.6545	0.5785	3.0986	3.3667	3.6063	3.8760	4.2357	2.1379	6.1054
Leverage	9,898	7.2881	3.0621	0.7966	6.3252	7.7454	9.0627	10.3153	0.0076	14.2192
Parent Debt Ratio	9,810	0.0187	0.0328	0.0000	0.0000	0.0084	0.0195	0.0543	0.0000	0.3356
TobinQ	9,898	1.0320	0.0540	0.9795	1.0000	1.0230	1.0531	1.0877	0.9122	1.5686
Ln TA	9,898	15.0257	2.2820	12.5406	13.9228	14.9970	16.3062	17.7974	7.2167	21.8888
HQLA Ratio	9,810	0.1542	0.0811	0.0664	0.0943	0.1391	0.1993	0.2605	0.0025	0.5890

Panel B: Reduced Sample (2012-2013 and 2016-2017)

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
Loan Lease Yield	4,172	0.0492	0.0160	0.0381	0.0420	0.0466	0.0522	0.0599	0.0184	0.136
Provision Ratio	4,172	0.00361	0.00681	-0.000356	0.000502	0.00192	0.00422	0.00930	-0.00993	0.0399
Z-Score	4,987	3.688	0.631	3.063	3.388	3.638	3.932	4.339	2.075	6.284
Z-Score Narrow	4,628	3.676	0.584	3.103	3.389	3.630	3.907	4.296	2.138	6.105
Leverage	4,643	7.346	3.078	1.459	6.352	7.834	9.078	10.44	0.00762	14.22
Parent Debt Ratio	4,991	0.0199	0.0340	0	0	0.00815	0.0245	0.0608	0	0.336
TobinQ	5,027	1.032	0.0538	0.979	0.998	1.024	1.056	1.089	0.927	1.502
Ln TA	5,026	14.78	2.315	12.28	13.72	14.68	16.04	17.62	0.693	21.49
HQLA Ratio	4,991	0.151	0.0776	0.0653	0.0929	0.138	0.196	0.256	0.00223	0.589

Panel C: Large Non-GSIB BHCs during the Pre-and Post-Period (2012-2013 and 2016-2017)

	2012-2103				2016-2017		
VARIABLES	N	mean	sd	p50	N	mean	sd	p50
HQLA Ratio	36	0.155	0.0440	0.156	36	0.206	0.0472	0.211
Z-Score_post	34	3.676	0.750	3.638	34	3.536	0.881	3.569
Z-Score_nrw_post	26	3.666	0.808	3.653	26	3.606	0.722	3.591
Provision Ratio	20	0.00801	0.0104	0.00324	20	0.00951	0.0150	0.00305
Loan Lease Yield	20	0.0543	0.0333	0.0435	20	0.0533	0.0359	0.0403
Leverage	26	4.984	3.334	6.386	26	5.491	3.559	6.788
TobinQ	35	1.048	0.0913	1.014	35	1.061	0.0700	1.042
Ln TA	36	15.15	3.862	16.17	36	15.33	4.113	16.98
Parent Debt Ratio	36	0.0297	0.0341	0.0279	36	0.0285	0.0227	0.0277
pdratiowda_ultp	36	0.0297	0.0341	0.0279	36	0.0285	0.0227	0.0278
pdratio_ultp	36	0.162	0.113	0.182	36	0.164	0.0774	0.189

Table 3

Parent debt ratio and subsidiary bank's risk-taking. This table presents the results from the model for our preliminary analysis used in Section (4). The dependent variables are the Loan Lease Yield, Provision Ratio, the logarithm of Z-score (Z-Score), and Leverage. The key explanatory variable is parent debt ratio. For each risk-taking measure, Models (1), (3), (5), (7) and (9) contain only parent debt ratio as a control, while Models (2), (4), (6), (8) and (10) contain all set of controls. All models use quarterly data for 2011-2021Q2. All models include both subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

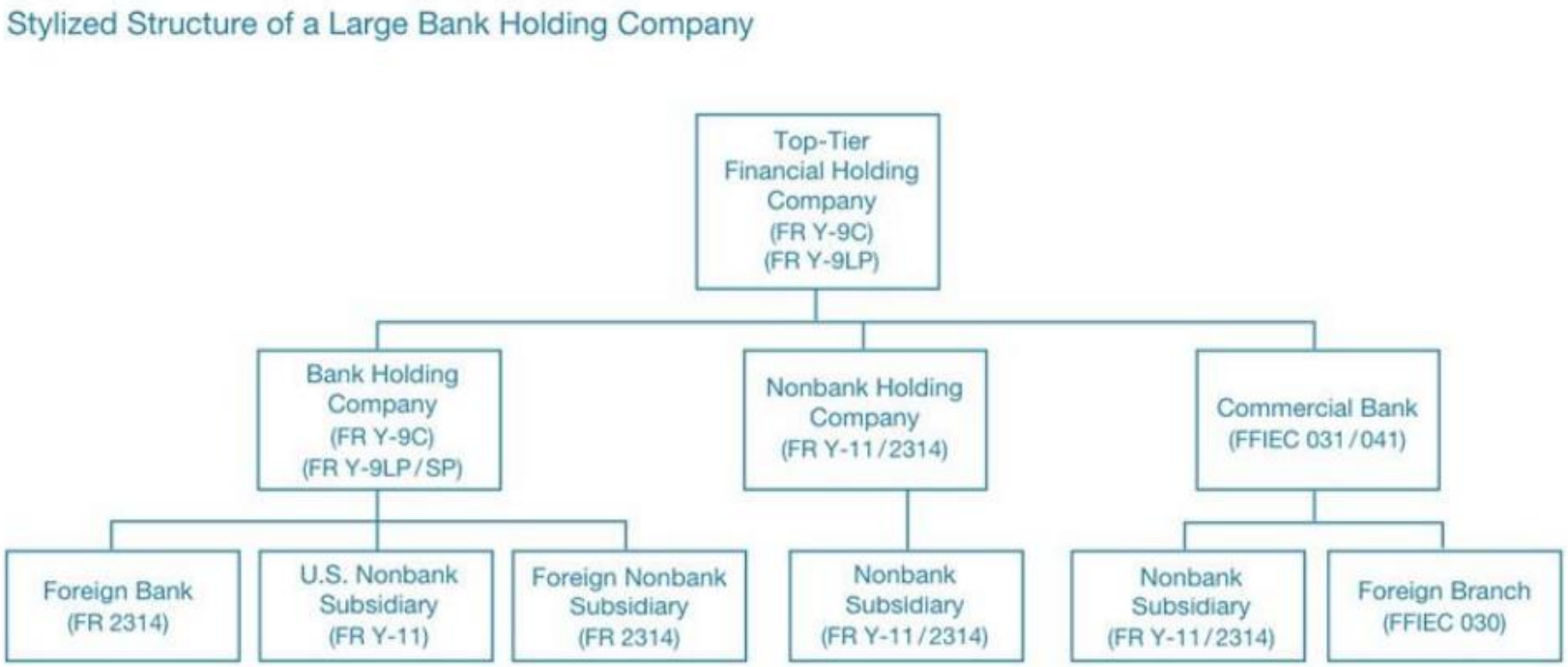
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	Loan Lease Yield	Loan Lease Yield	Provision Ratio	Provision Ratio	Z-Score	Z-Score	Z-Score Narrow	Z-Score Narrow	Leverage	Leverage
Parent Debt Ratio	-0.068**	-0.062*	-0.047***	-0.048***	3.966***	2.967***	3.951***	2.973***	-17.295***	-16.905***
	(0.035)	(0.037)	(0.017)	(0.018)	(1.364)	(1.132)	(1.335)	(1.122)	(4.655)	(4.413)
Ln TA		-0.001		0.000		-0.156*		-0.153*		0.041
		(0.001)		(0.000)		(0.092)		(0.089)		(0.272)
Leverage		0.000		0.000		-0.054***		-0.053***		
		(0.000)		(0.000)		(0.013)		(0.012)		
HQLA Ratio		-0.001		-0.008**		0.188		0.147		4.314***
		(0.007)		(0.003)		(0.243)		(0.244)		(0.957)
TobinQ		0.013		-0.004		-0.352		-0.400		-0.392
		(0.010)		(0.004)		(0.336)		(0.336)		(1.190)
Constant	0.049***	0.052***	0.004***	0.002	3.579***	6.662***	3.582***	6.659***	7.608***	6.744
	(0.001)	(0.017)	(0.000)	(0.008)	(0.025)	(1.611)	(0.025)	(1.556)	(0.086)	(4.492)
Observations	8,907	8,907	8,907	8,907	9,871	9,871	9,871	9,871	9,898	9,898
R-squared	0.869	0.870	0.571	0.573	0.655	0.671	0.653	0.669	0.878	0.881
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 4

Difference-in-Differences Analysis in Subsidiary Bank's Risk-taking. This table reports results from difference-in-differences regressions using Equation (2). The treated group includes BHCs with total consolidated assets equal to or greater than 50 billion dollars as of 2014Q1 but excludes GSIBs, while the control group consists of BHCs with total consolidated assets less than 50 billion dollars as of 2014Q1. Post is an indicator variable equal to zero before 2014 and one after 2015. All columns collapse each pre- and post-period into a single observation for each subsidiary bank. Models (1), (3), (5), (7) and (9) contain Nonbank and Time as controls, while Models (2), (4), (6), (8) and (10) contain all set of controls. All models include subsidiary bank fixed effect. T-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	Loan Lease Yield	Loan Lease Yield	Provision Ratio	Provision Ratio	Z-Score	Z-Score	Z-Score Narrow	Z-Score Narrow	Leverage	Leverage
Post	-0.007***	-0.008***	-0.002***	-0.002***	0.096***	0.139*	0.098***	0.254***	-0.347***	-0.399**
	(0.000)	(0.001)	(0.000)	(0.001)	(0.031)	(0.080)	(0.031)	(0.072)	(0.075)	(0.168)
TreatBHC x Post	0.006***	0.006***	0.003***	0.003**	-0.225*	-0.244**	-0.139**	-0.236***	0.854***	0.791***
	(0.002)	(0.002)	(0.001)	(0.001)	(0.121)	(0.122)	(0.065)	(0.086)	(0.234)	(0.261)
GSIBs x Post	-0.004	-0.004	-0.001	0.001	-0.138	-0.108	-0.100	-0.181	0.230	0.143
	(0.005)	(0.005)	(0.002)	(0.002)	(0.128)	(0.126)	(0.121)	(0.134)	(0.304)	(0.287)
TobinQ		0.016		-0.029***		-3.013**		-3.243***		2.989
		(0.019)		(0.009)		(1.256)		(1.241)		(2.870)
Ln TA		0.001		0.002***		0.128		-0.108		0.004
		(0.002)		(0.001)		(0.124)		(0.093)		(0.272)
Leverage		-0.000		0.001***						
		(0.001)		(0.000)						
HQLA Ratio		0.008		-0.021***		-0.196		0.050		1.809
		(0.012)		(0.007)		(0.604)		(0.603)		(1.779)
Constant	0.053***	0.022	0.004***	-0.003	3.606***	4.846**	3.602***	8.475***	7.782***	4.395
	(0.000)	(0.046)	(0.000)	(0.017)	(0.015)	(2.298)	(0.014)	(1.998)	(0.035)	(5.005)
Observations	634	634	634	634	728	728	682	682	690	690
R-squared	0.936	0.937	0.792	0.817	0.741	0.750	0.764	0.772	0.947	0.947
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Figure 1: The organizational structure of a large BHC



Data sources: Avraham et al. (2012)

Figure 2: Parent debt ratio over time

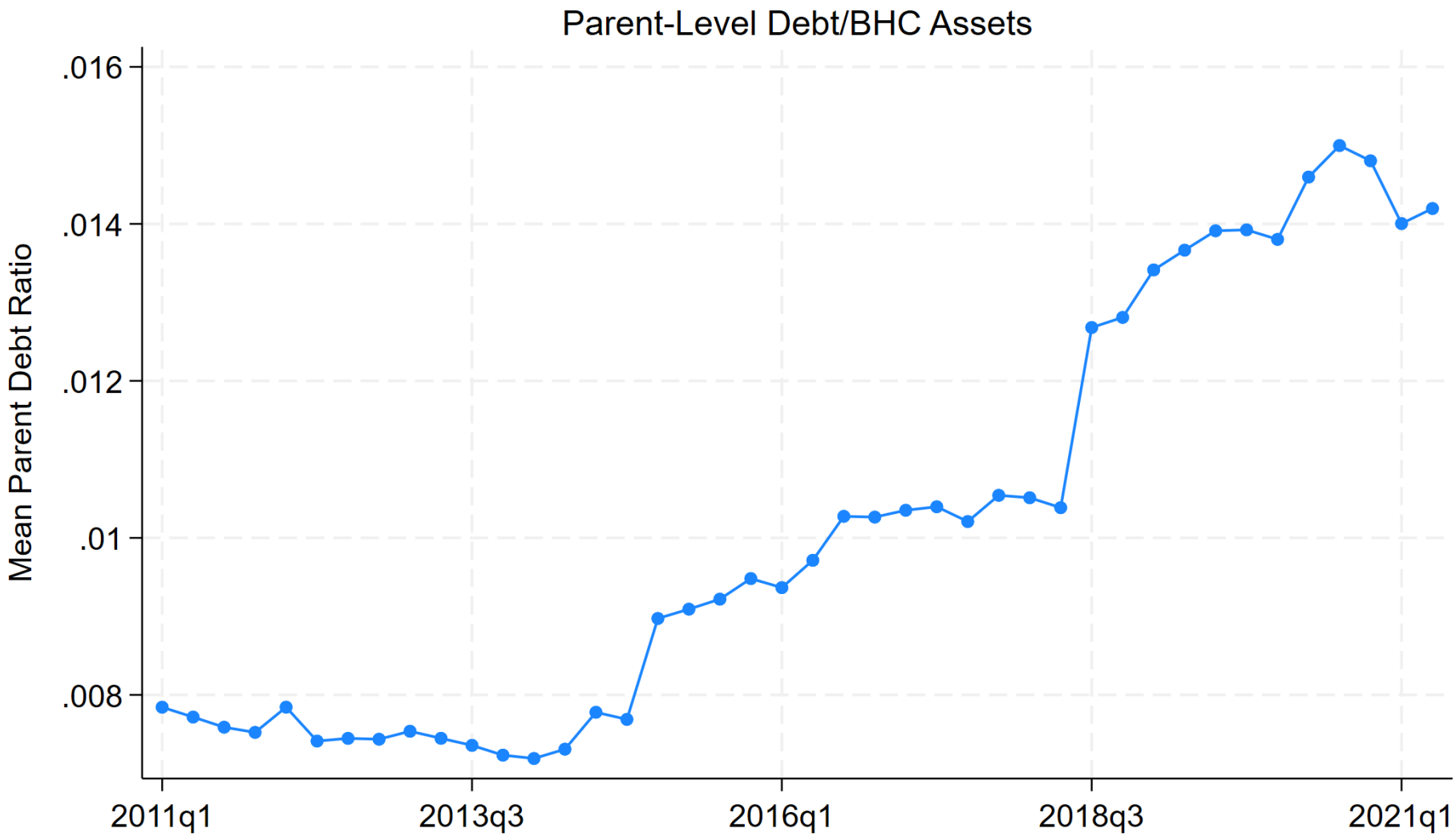


Figure 3: Subsidiary bank's asset ratio over time

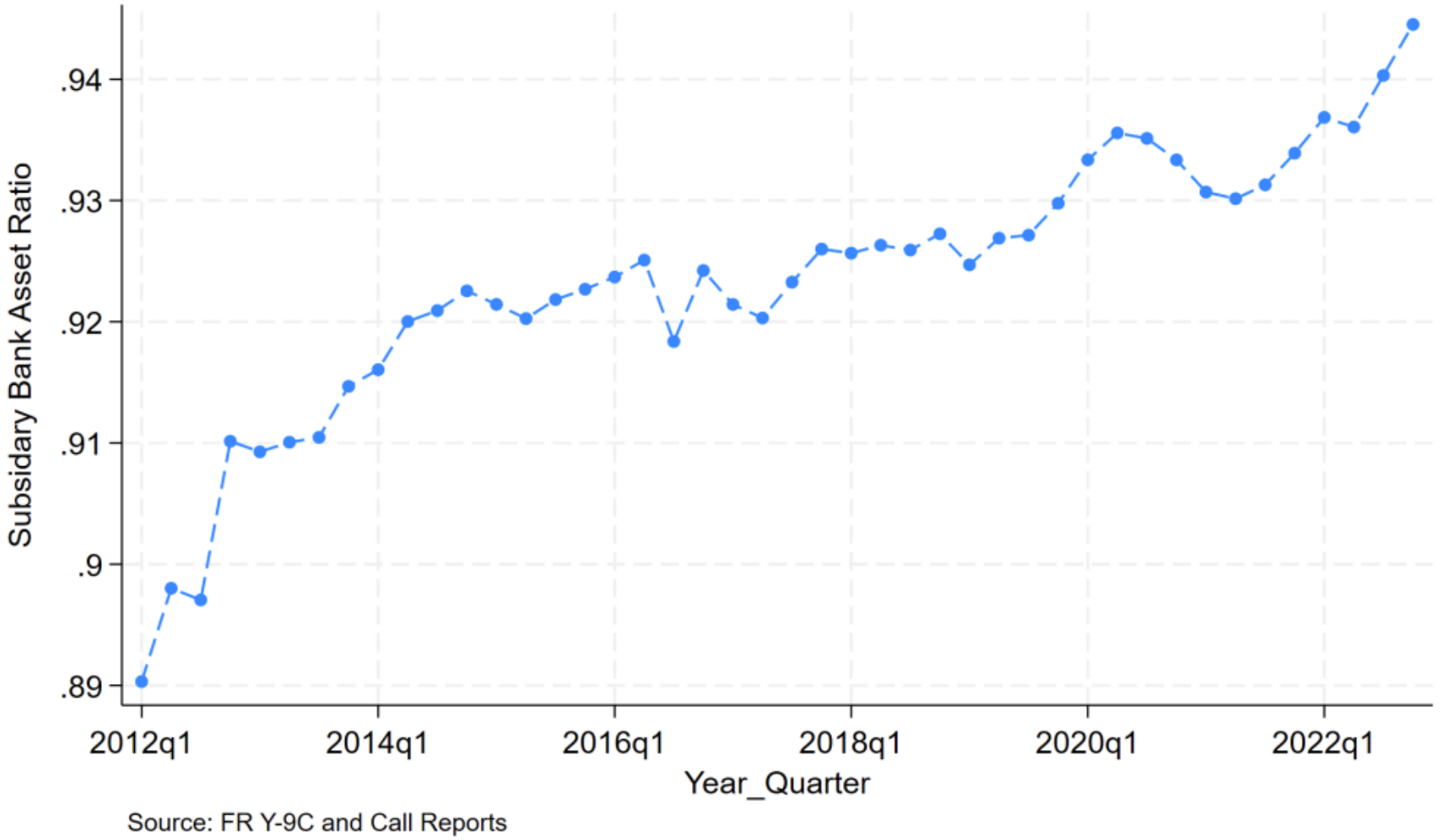


Figure 4: Parallel trend for large non-GSIB BHCs and the control group

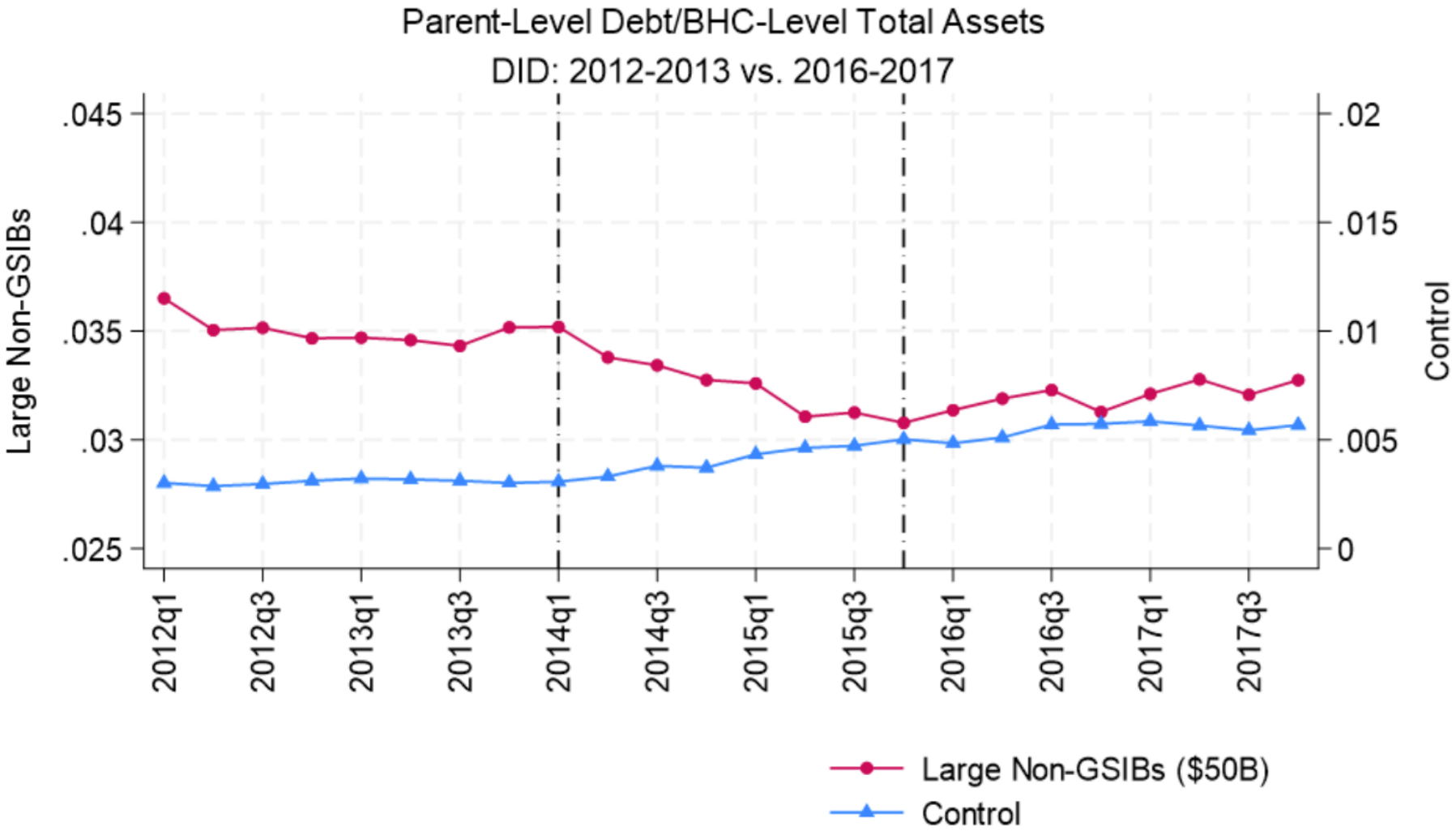


Figure 5: Parallel trend for GSIBs and the control group

